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DISPLAY PANEL, DISPLAY APPARATUS, AND METHOD FOR MANUFACTURING DISPLAY PANEL

Abstract

A display panel, a display apparatus, and a method for manufacturing a display panel. The display panel includes a base plate, a plurality of isolation structures, and a light-emitting layer. A first layer and a second layer are arranged to form the isolation structure, and the orthographic projection of the first layer arranged close to the base plate on the base plate is located within the orthographic projection of the second layer on the base plate.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Chinese Patent Application No. 202410232931.8 filed on Feb. 29, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present application relates to the field of display technology, and particularly to a display panel, a display apparatus, and a method for manufacturing a display panel.

BACKGROUND

[0003] Planar display apparatus based on Organic Light Emitting Diode (OLED) and Light Emitting Diode (LED), etc., are widely used in cell phones, TVs, notebook computers, desktop computers and other consumer electronic products due to their high image quality, power saving, thin body and wide range of applications, and have become the mainstream of the display apparatus.

[0004] However, the usability of the current OLED display products needs to be improved.

SUMMARY

[0005] Embodiments of the present application provide a display panel, a display apparatus, and a method for manufacturing a display panel, aiming to improve the usability of the display panel.

[0006] Some embodiments of a first aspect of the present application provide a display panel, including: a base plate; a plurality of isolation structures located on the base plate, the isolation structure including a first layer and a second layer located at a side of the first layer away from the base plate, an orthographic projection of the first layer on the base plate being located within an orthographic projection of the second layer on the base plate; a plurality of isolation openings encircled by the isolation structures; and a light-emitting layer located on the base plate and including light-emitting units located in the isolation openings; in which a distance between an edge of the orthographic projection of the first layer on the base plate close to the light-emitting unit and an edge of the orthographic projection of the second layer on the base plate close to the light-emitting unit is an extension distance, and in at least two of the light-emitting units, the extension distance of the isolation structure close to one of the two light-emitting units is greater than the extension distance of the isolation structure close to the other one of the two light-emitting units.

[0007] Some embodiments of a second aspect of the present application provide a display panel, including: a base plate; a plurality of isolation structures located on the base plate, the isolation structure including a first layer and a second layer located at a side of the first layer away from the base plate, an orthographic projection of the first layer on the base plate being located within an orthographic projection of the second layer on the base plate; a plurality of isolation openings encircled by the isolation structures; and a light-emitting layer located on the base plate and including light-emitting units located in the isolation openings; in which the first layer includes a top surface at a side away from the base plate, the second layer includes a bottom surface at a side close to the base plate, the top surface and the bottom surface are in contact, a distance between an edge of an orthographic projection of the top surface on the base plate close to the light-emitting unit and an edge of an orthographic projection of the bottom surface on the base plate close to the light-emitting unit is an extension distance, and in at least two of the light-emitting units, the extension distance of the isolation structure close to one of the two light-emitting units is greater than the extension distance of the isolation structure close to the other one of the two light-emitting units, wherein the extension distances close to the light-emitting units of a same color are same; and the extension distances close to the light-emitting units of different colors are different.

[0008] Some embodiments of a third aspect of the present application provide a display panel, including: a base plate; a plurality of isolation structures located on the base plate, the isolation structure including a first layer and a second layer located at a side of the first layer away from the base plate, an orthographic projection of the first layer on the base plate being located within an orthographic projection of the second layer on the base plate; a plurality of isolation openings encircled by the isolation structures; and a light-emitting layer located on the base plate and including light-emitting units located in the isolation openings; in which a distance between orthographic projections of the first layers located at two sides of a same isolation opening on the base plate is a sixth distance, and in at least two of the light-emitting units, the sixth distance of the isolation structure close to one of the two light-emitting units is greater than the sixth distance of the isolation structure close to the other one of the two light-emitting units.

[0009] Some embodiments of a fourth aspect of the present application provide a display apparatus, including the display panel according to any of the above implementations.

[0010] Some embodiments of a fifth aspect of the present application provide a method for manufacturing a display panel, including: preparing a plurality of isolation structures on a base plate, a first isolation opening and a second isolation opening encircled by the isolation structures, the isolation structure including a first layer and a second layer located at a side of the first layer away from the base plate, an orthographic projection of the first layer on the base plate being located within an orthographic projection of the second layer on the base plate, a distance between an edge of the orthographic projection of the first layer on the base plate close to a light-emitting unit and an edge of the orthographic projection of the second layer on the base plate close to the light-emitting unit being an extension distance; preparing a first light-emitting material layer and a first electrode material layer on the base plate and patterning the first light-emitting material layer and the first electrode material layer to form a first light-emitting unit and a first sub-electrode located in the first isolation opening, the extension distance including a first distance close to the first light-emitting unit; and preparing a second light-emitting material layer and a second electrode material layer on the base plate and patterning the second light-emitting material layer and the second electrode material layer to form a second light-emitting unit and a second sub-electrode located in the second isolation opening, the extension distance including a second distance close to the second light-emitting unit, and the second distance being greater than the first distance.

[0011] The display panel according to the embodiments of the present application includes the base plate, the isolation structures, and the light-emitting layer. The first layer and the second layer are arranged to form the isolation structure, the orthographic projection of the first layer arranged close to the base plate on the base plate is located within the orthographic projection of the second layer on the base plate, the area of the second layer is greater than the area of the first layer, the second layer covers a surface of the first layer close to the second layer, and in such cases, the first layer is recessed with respect to the second layer in a direction away from the isolation opening. When the light-emitting layer is prepared, the light-emitting layer has a large drop at the edge of the isolation structure, and the first layer is concave with respect to the second layer, it's difficult for the light-emitting layer to be connected at the edge of the isolation structure, and thus fracture occurs and the light-emitting layer is fractured, resulting in the light-emitting units that are spaced apart from each other. In this way, the crosstalk of carriers within the light-emitting layer is reduced, the display effect of the display panel is improved, and the light-emitting units can be prepared without a precision mask plate, so as to reduce the development and use of the precision mask plate and reduce the preparation cost.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Other features, objects and advantages of the present application will be more apparent from reading the following detailed description of non-limiting embodiments with reference to the accompanying drawings, in which the same or similar reference numerals denote the same or similar features, and the accompanying drawings are not drawn to actual scale.

[0013] FIG. **1** shows a partial sectional view of a display panel according to an embodiment of the present application;

[0014] FIG. **2** shows a partial sectional view of a display panel in another embodiment;

[0015] FIG. **3** shows a partial sectional view of a display panel in yet another embodiment;

[0016] FIG. **4** shows a partial top view of a display panel according to an embodiment of the present application;

[0017] FIG. **5** shows a partial sectional view of a display panel in yet another embodiment;

[0018] FIG. **6** shows a partial sectional view of a display panel in yet another embodiment;

[0019] FIG. **7** shows a partial sectional view of a display panel in yet another embodiment;

[0020] FIG. **8** shows a partial sectional view of a display panel in yet another embodiment;

[0021] FIG. **9** shows a partial sectional view of a display panel in yet another embodiment;

[0022] FIG. **10** shows a partial sectional view of a display panel in yet another embodiment;

[0023] FIG. **11** shows a partial top view of a display panel in yet another embodiment;

[0024] FIG. **12** shows a schematic flow chart of a method for manufacturing a display panel according to an embodiment of the present application;

[0025] FIGS. **13** to **16** show diagrams of a process for manufacturing a display panel according to an embodiment of the present application.

REFERENCE NUMERALS

[0026] **10**, display panel; [0027] **100**, base plate; [0028] **200**, isolation structure; **210**, first layer; **211**, top surface; **220**, second layer; **221**, bottom surface; **230**, third layer; **240**, isolation opening; **241**, first isolation opening; **242**, second isolation opening; **243**, third isolation opening; [0029] **300**, light-emitting layer; **310**, light-emitting unit; **311**, first light-emitting unit; **312**, second light-emitting unit; **313**, third light-emitting unit; [0030] **400**, first electrode layer; **410**, first electrode; **411**, first sub-electrode; **412**, second sub-electrode; **413**, third sub-electrode; [0031] **500**, pixel defining layer; **510**, pixel defining portion; **520**, pixel opening; **530**, pixel electrode; [0032] D**0**, extension distance; D**1**, first distance; D**2**, second distance; D**3**, third distance; D**4**, fourth distance; D**5**, fifth distance; D**6**, sixth distance; D**7**, seventh distance; D**8**, eighth distance; D**9**, ninth distance.

DETAILED DESCRIPTION

[0033] Features and exemplary embodiments of various aspects of the present application will be described in detail below. In order to make the objectives, technical solutions, and advantages of the present application clearer, the present application will be further described in detail below with reference to the drawings and specific embodiments. It should be understood that the specific embodiments described herein are merely configured to explain the present application, rather than to limit the present application. For those skilled in the art, the present application can be implemented without some of these specific details. The following description of the embodiments is merely to provide a better understanding of the present application by illustrating the examples of the present application.

[0034] It should be noted that, in the present application, relational terms, such as first and second, are used merely to distinguish one entity or operation from another entity or operation, without necessarily requiring or implying any actual such relationships or orders for these entities or operations. Moreover, the terms “comprise”, “include”, or any other variants thereof, are intended to represent a non-exclusive inclusion, such that a process, method, article or device comprising/including a series of elements includes not only those elements, but also other elements that are not explicitly listed or elements inherent to such a process, method, article or device. Without more constraints, the elements following an expression "comprise/include . . . " do not exclude the existence of additional identical elements in the process, method, article or device that includes the elements.

[0035] It should be understood that when describing the structure of a component, if a layer/area is referred to as being "on" or "above" another layer/region, it may mean that the layer/area is directly on the other layer/region or that other layers/regions may be included between the layer/area and the other layer/area. Moreover, if the component is turned over, the layer/region will be "below" or "under" the other layer/region.

[0036] The embodiments of the present application provide a display panel, a display apparatus, and a method for manufacturing a display panel, and various embodiments of the display panel, the display apparatus, and the method for manufacturing a display panel will be described below in conjunction with the accompanying drawings.

[0037] The embodiments of the present application provide a display panel which may be an Organic Light Emitting Diode (OLED) display panel.

[0038] Reference is made to FIG. **1**, which shows a partial sectional view of a display panel according to an embodiment of the present application.

[0039] As shown in FIG. **1**, the embodiments of the first aspect of the present application provide a display panel **10**, including: a base plate **100**; a plurality of isolation structures **200** located on the base plate **100**, the isolation structures **200** encircling a plurality of isolation openings **240**, the isolation structure **200** including a first layer **210** and a second layer **220** located at a side of the first layer **210** away from the base plate **100**, an orthographic projection of the first layer **210** on the base plate **100** being located within an orthographic projection of the second layer **220** on the base plate **100**; and a light-emitting layer **300** located on the base plate **100** and including light-emitting units **310** located in the isolation openings **240**; in which a distance between an edge of the orthographic projection of the first layer **210** on the base plate **100** close to the light-emitting unit **310** and an edge of the orthographic projection of the second layer **220** on the base plate **100** close to the light-emitting unit **310** is an extension distance DO, and in at least two of the light-emitting units **310**, the extension distance D**0** of the isolation structure **200** close to one of the two light-emitting units **310** is greater than the extension distance D**0** of the isolation structure **200** close to the other one of the two light-emitting units **310**.

[0040] The display panel **10** according to the embodiments of the present application includes the base plate **100**, the isolation structures **200**, and the light-emitting layer **300**. The first layer **210** and the second layer **220** are arranged to form the isolation structure **200**, the orthographic projection of the first layer **210** arranged close to the base plate **100** on the base plate **100** is located within the orthographic projection of the second layer **220** on the base plate **100**, the area of the second layer **220** is greater than the area of the first layer **210**, the second layer **200** covers a surface of the first layer **210** close to the second layer **220**, and in such cases, the first layer **210** is recessed with respect to the second layer **220** in a direction away from the isolation opening **240**. When the light-emitting layer **300** is prepared, the light-emitting layer **300** has a large drop at the edge of the isolation structure **200**, and the first layer **210** is concave with respect to the second layer **220**, it's difficult for the light-emitting layer **300** to be connected at the edge of the isolation structure **200**, and thus fracture occurs and the light-emitting layer **300** is fractured, resulting in the light-emitting units **310** that are spaced apart from each other. In this way, the crosstalk of carriers within the light-emitting layer **300** is reduced, the display effect of the display panel **10** is improved, and the light-emitting units **310** can be prepared without a precision mask plate, so as to reduce the development and use of the precision mask plate and reduce the preparation cost. For example, the light-emitting units **310** include a blue light-emitting unit, a green light-emitting unit, and a red light-emitting unit, the extension distance D**0** of the isolation structure **200** close to the blue light-emitting unit is greater than the extension distance D**0** of the isolation structure **200** close to the green light-emitting unit, and the extension distance D**0** of the isolation structure **200** close to the green light-emitting unit is greater than the extension distance D**0** of the isolation structure **200** close to the red light-emitting unit, and thus it may be determined that the red light-emitting unit is prepared first, followed by the green light-emitting unit, and finally the blue light-emitting unit.

[0041] The extension distance D**0** of the isolation structure **200** close to the light-emitting unit **310** refers to a distance between an edge of the first layer **210** of the isolation structure **200**, which is located at the periphery of the light-emitting unit **310** and surrounding the light-emitting unit **310**, close to the light-emitting unit **310** and an edge of a contact surface between the second layer **220** and the first layer **210** close to the light-emitting unit **310**.

[0042] The base plate **100** may be arranged in various ways. For example, the base plate **100** may include a substrate and an array base plate arranged on the substrate. Alternatively, the base plate **100** is the substrate. Alternatively, the base plate **100** includes a buffer layer, a support plate, and the like at a side away from the base plate.

[0043] Related solutions of the isolation structure are described in patent applications No. PCT/CN2023/134518, No. 202310759370.2, No. 202311117143.6, No. 202311499823.9, No. 202310707209.0, No. 202311346196.5, No. 202310692671.8, and No. 202310909421.5, the contents of which are incorporated herein by reference and will not be repeated herein.

[0044] In some optional embodiments, the extension distances D**0** close to the light-emitting units **310** of a same color are the same.

[0045] In these optional embodiments, the extension distances D**0** of the isolation structures **200** located at the periphery of the light-emitting units **310** of a same color are the same, it may be determined that the light-emitting units **310** of a same color are prepared in a same step. Moreover, the extension distances D**0** corresponding to the light-emitting units **310** of a same color are the same, and if the extension distances D**0** corresponding to at least two light-emitting units **310** are different, the colors of lights emitted by the two light-emitting units **310** are different, and the two light-emitting units **310** are vapor deposited in different steps. Therefore, the extension distances D**0** corresponding to the light-emitting units **310** of a same color being controlled to be the same facilitates determining the vapor deposition order for the light-emitting units **310** according to the extension distances D**0** corresponding to the light-emitting units **310**.

[0046] In some optional embodiments, the extension distances D**0** close to the light-emitting units **310** of different colors are different.

[0047] In these optional embodiments, the extension distances D**0** of the isolation structures **200** located at the periphery of the light-emitting units **310** of different colors are different, it may be determined that the light-emitting units **310** of different colors are prepared in different steps. The extension distances D**0** corresponding to the light-emitting units **310** of a same color are the same, and the extension distances D**0** corresponding to the light-emitting units **310** of different colors are different, therefore the vapor deposition order for the light-emitting units **310** can be determined according to the extension distances D**0** corresponding to the light-emitting units **310**.

[0048] As shown in FIG. **1**, in some optional embodiments, the first layer **210** includes a top surface **211** at a side away from the base plate **100**, the second layer **220** includes a bottom surface **221** at a side close to the base plate **100**, the top surface **211** and the bottom surface **221** are in contact, a distance between an edge of an orthographic projection of the top surface **211** on the base plate **100** close to the light-emitting unit **310** and an edge of an orthographic projection of the bottom surface **221** on the base plate **100** close to the light-emitting unit **310** is the extension distance DO.

[0049] In these optional embodiments, in the preparation process, the isolation structures **200** may be wet etched in a wet etching process when the light-emitting units **310** of different colors are prepared, i.e., the first layers **210** of the isolation structures **200** may be etched. Therefore, in order to reflect the difference in the etching of the first layers **210**, the distance between the edge of the orthographic projection of the top surface **211** on the base plate **100** close to the light-emitting unit **310** and the edge of the orthographic projection of the bottom surface **221** on the base plate **100** close to the light-emitting unit **310** is used as the extension distance DO, so as to determine the vapor deposition order for the light-emitting units **310** according to the extension distances D**0** corresponding to the light-emitting units **310**.

[0050] Reference is made to FIGS. **2** to **4**, in which FIG. **2** shows a partial sectional view of a display panel in another embodiment, FIG. **3** shows a partial sectional view of a display panel in yet another embodiment, and FIG. **4** shows a partial top view of a display panel according to an embodiment of the present application.

[0051] As shown in FIGS. **2** to **4**, in some optional embodiments, the light-emitting units **310** include a first light-emitting unit **311**, a second light-emitting unit **312**, and a third light-emitting unit **313** of different colors, the extension distance D**0** of the isolation structure **200** close to the first light-emitting unit **311** is a first distance D**1**, the extension distance D**0** of the isolation structure **200** close to the second light-emitting unit **312** is a second distance D**2**, the extension distance D**0** of the isolation structure **200** close to the third light-emitting unit **313** is a third distance D**3**, and at least two of the first distance D**1**, the second distance D**2**, and the third distance D**3** are different.

[0052] In these optional embodiments, at least two of the first distance D**1**, the second distance D**2**, and the third distance D**3** are different, i.e., the extension distances DO corresponding to at least two of the first light-emitting unit **311**, the second light-emitting unit **312**, and the third light-emitting unit **313** are different, and the vapor deposition order for the light-emitting units may be determined according to the different extension distances DO. For example, the third distance D**3** is greater than the second distance D**2**, i.e., the extension distance D**0** of the isolation structure **200** close to the third light-emitting unit **313** is greater than the extension distance D**0** of the isolation structure **200** close to the second light-emitting unit **312**, it may be determined that the second light-emitting unit **312** is prepared first, followed by the third light-emitting unit **313**; the second distance D**2** is greater than the first distance D**1**, i.e., the extension distance D**0** of the isolation structure **200** close to the second light-emitting unit **312** is greater than the extension distance D**0** of the isolation structure **200** close to the first light-emitting unit **311**, it may be determined that the first light-emitting unit **311** is prepared first, followed by the second light-emitting unit **312**.

[0053] Optionally, the first light-emitting unit **311** is a light-emitting unit emitting red light, the second light-emitting unit **312** is a light-emitting unit emitting green light, and the third light-emitting unit **313** is a light-emitting unit emitting blue light.

[0054] In some optional embodiments, the extension distance D**0** is in a range of 0.1 μm to 1.5 μm.

[0055] In these optional embodiments, the extension distance D**0** is greater than or equal to 0.1 μm to alleviate the following problem: the extension distance D**0** is too small, and thus the concavity of the first layer **210** with respect to the second layer **220** is too less, it is difficult for the light-emitting layer **300** to be disconnected at the edge of the second layer **220**, and the light-emitting layer **300** is likely to be in contact with the first layer **210** after being disconnected, causing crosstalk of carriers within the light-emitting layer **300**. The extension distance D**0** is less than or equal to 1.5 μm to alleviate the following problem: the extension distance D**0** is too large, and thus the concavity of the first layer **210** with respect to the second layer **220** is too much, the first electrode **410** is deposited within the isolation opening **240** far away from the sidewall of the first layer **210**, and it is difficult for the first electrode **410** to overlap the first layer **210**, causing poor overlapping.

[0056] As shown in FIGS. **1** to **3**, in some optional embodiments, the display panel **10** further includes a first electrode layer **400** located a side of the light-emitting layer **300** away from the base plate **100**.

[0057] Optionally, the first electrode layer **400** includes a plurality of first electrodes **410** arranged at intervals, the first electrode **410** is electrically connected with the isolation structure **200**.

[0058] In these optional embodiments, the isolation structures **200** separate the first electrode layer **400** to form the first electrodes **410** that are spaced apart from each other, and the first electrodes **410** are electrically connected through the isolation structures **200** to form a planar electrode, so as to ensure normal light emission of the light-emitting unit **310**.
[0059] In some optional embodiments, an orthographic projection of the light-emitting unit **310** on the base plate **100** is located within an orthographic projection of the first electrode **410** on the base plate **100**.
[0060] In these optional embodiments, the orthographic projection of the light-emitting unit **310** on the base plate **100** is located within the orthographic projection of the first electrode **410** on the base plate **100**, i.e., the first electrode **410** is arranged covering the light-emitting unit **310** to be used as an electrode of the light emitting-unit **310**, so as to ensure normal light emission of the light-emitting unit **310** and improve the display effect of the display panel **10**.
[0061] Optionally, the light-emitting units **310** and the isolation structures **200** are arranged alternately, i.e., the light-emitting units **310** are spaced apart from each other, so as to reduce crosstalk of carriers and color crossing among the light-emitting units **310**.
[0062] In some optional embodiments, the second layer **220** includes an electrically conductive material or an electrically insulating material.
[0063] In these optional embodiments, the second layer **220** includes an electrically conductive material, such as a non-metal electrically conductive material or a metal electrically conductive material. If the second layer **220** includes a non-metal electrically conductive material or an electrically insulating material, it is difficult to etch the second layer **220** during the wet etching of the first layer **210** using an etching solution, thereby making it easier to achieve the concavity of the first layer **210** with respect to the second layer **220**.
[0064] In some optional embodiments, the second layer **220** includes a metal material, and the first layer **210** and the second layer **220** are of different materials.
[0065] In these optional embodiments, if both the first layer **210** and the second layer **220** include metal materials, the first layer **210** may be wet etched using an etching solution, and the etching rate of the second layer **220** may be less than the etching rate of the first layer **210** by selecting the etching solution. Since the etching rate of the first layer **210** is greater, when the first layer **210** is wet etched using the etching solution, even though the second layer **220** may be etched to some degree, the first layer **210** is etched faster, and thus the first layer **210** is concave with respect to the second layer **220**.
[0066] Reference is made to FIG. **5**, which shows a partial sectional view of a display panel in yet another embodiment.
[0067] As shown in FIG. **5**, in some optional embodiments, the isolation structure **200** further includes a third layer **230** located at a side of the first layer **210** towards the base plate **100**, the orthographic projection of the first layer **210** on the base plate **100** is located within an orthographic projection of the third layer **230** on the base plate **100**.
[0068] In these optional embodiments, in order to obtain the concave first layer **210**, the first layer **210** has a greater etching rate than the second layer **220** and the third layer **230** during the etching, thereby forming the concave first layer **210**. Due to the greater etching rate of the first layer **210**, a great amount of waste materials produced by the etching are likely to get into other locations of the display panel **10**, causing undesirable effects. After the third layer **230** being arranged, the first layer **210** can be better attached to the third layer **230**, and the waste materials produced by the etching will fall on the third layer **230**, which are easy to be cleaned up.
[0069] Optionally, the second layer **220** is made of titanium (Ti), the first layer **210** is made of aluminum (Al), and the third layer **230** is made of titanium (Ti) or molybdenum (Mo), i.e., the isolation structure **200** includes a triple-layer metal composite material such as Ti/Al/Ti (titanium/aluminum/titanium) or Ti/Al/Mo (titanium/aluminum/molybdenum).
[0070] In some optional embodiments, the display panel **10** further includes a pixel defining layer

500 located on the base plate **100**, the pixel defining layer **500** includes a plurality of pixel defining portions **510** and a plurality of pixel openings **520** encircled by the pixel defining portions **510**, the pixel opening **520** is communicated with the isolation opening **240**.

[0071] In these optional embodiments, the pixel defining portions **510** encircle the pixel openings **520** to define the light-emitting area of the display panel **10**. The pixel opening **520** is communicated with the isolation opening **240** to reduce the blocking of the pixel opening **520** by the isolation structure **200**, so as to ensure the light emitting effect of the display panel **10**.

[0072] In some optional embodiments, the isolation structure **200** is located at a side of the pixel defining portion **510** away from the base plate **100**.

[0073] In these optional embodiments, the isolation structure **200** is arranged on the pixel defining portion **510** and has a large height drop with respect to the pixel opening **520**. When the first electrode layer **400** is prepared, the first electrode layer **400** is more likely to be disconnected at the isolation structure **200** due to the large drop, reducing the difficulty for preparing the first electrode layer **400**.

[0074] In some optional embodiments, the display panel **10** further includes a plurality of pixel electrodes **530** located at a side of the pixel defining portions **510** close to the base plate **100**, an orthographic projection of the pixel electrode **530** on the base plate **100** at least partially overlaps an orthographic projection of the pixel opening **520** on the base plate **100**.

[0075] In these optional embodiments, the orthographic projection of the pixel electrode **530** on the base plate **100** at least partially overlaps the orthographic projection of the pixel opening **520** on the base plate **100**, i.e., at least a portion of the pixel electrode **530** is exposed from the pixel opening **520** to be used as an electrode of the light-emitting unit **310**, so as to ensure the light emission of the light-emitting unit **310**. One of the pixel electrode **530** and the first electrode **410** is used as the anode of the light-emitting unit **310**, and the other one is used as the cathode of the light-emitting unit **310**. In the embodiments of the present application, for example, the pixel electrode **530** is used as the anode of the light-emitting unit **310**, and the first electrode **410** is used as the cathode of the light-emitting unit **310**.

[0076] Optionally, an orthographic projection of the isolation structure **200** on the base plate **100** is located within an orthographic projection of the pixel defining portion **510** on the base plate **100**, and the isolation structure **200** is located entirely on the pixel defining portion **510**, so as to reduce the blocking of the pixel opening **520** by the isolation structure **200**, thereby reducing the blocking of light emission of the light-emitting unit **310** by the isolation structure **200**, and ensuring the light emitting effect of the light-emitting unit **310**.

[0077] Reference is made to FIG. **6**, which shows a partial sectional view of a display panel in yet another embodiment.

[0078] As shown in FIG. **6**, optionally, a distance between the edge of the orthographic projection of the first layer **210** on the base plate **100** close to the light-emitting unit **310** and an edge of the orthographic projection of the pixel opening **520** on the base plate **100** is a fourth distance D**4**, and in at least two of the light-emitting units **310**, the fourth distance D**4** of the isolation structure **200** close to one of the two light-emitting units **310** is greater than the fourth distance D**4** of the isolation structure **200** close to the other one of the two light-emitting units **310**. For example, the light-emitting units **310** include a blue light-emitting unit, a green light-emitting unit, and a red light-emitting unit, the fourth distance D**4** of the isolation structure **200** close to the blue light-emitting unit is greater than the fourth distance D**4** of the isolation structure **200** close to the green light-emitting unit, and the fourth distance D**4** of the isolation structure **200** close to the green light-emitting unit is greater than the fourth distance D**4** of the isolation structure **200** close to the red light-emitting unit, and thus it may be determined that the red light-emitting unit is prepared first, followed by the green light-emitting unit, and finally the blue light-emitting unit.

[0079] Optionally, the fourth distances D**4** close to the light-emitting units **310** of a same color are the same.

[0080] In these optional embodiments, the fourth distances D**4** of the isolation structures **200** located at the periphery of the light-emitting units **310** of a same color are the same, it may be determined that the light-emitting units **310** of a same color are prepared in a same step. Moreover, the fourth distances D**4** corresponding to the light-emitting units **310** of a same color are the same, and if the fourth distances D**4** corresponding to at least two light-emitting units **310** are different, the colors of lights emitted by the two light-emitting units **310** are different, and the two light-emitting units **310** are vapor deposited in different steps. Therefore, the fourth distances D**4** corresponding to the light-emitting units **310** of a same color being controlled to be the same facilitates determining the vapor deposition order for the light-emitting units **310** according to the fourth distances D**4** corresponding to the light-emitting units **310**.
[0081] Reference is made to FIG. **7**, which shows a partial sectional view of a display panel in yet another embodiment.
[0082] As shown in FIG. **7**, optionally, a distance between an edge of the orthographic projection of the third layer **230** on the base plate **100** close to the light-emitting unit **310** and the edge of the orthographic projection of the second layer **220** on the base plate **100** close to the light-emitting unit **310** is a fifth distance D**5**, and in at least two of the light-emitting units **310**, the fifth distance D**5** of the isolation structure **200** close to one of the two light-emitting units **310** is greater than the fifth distance D**5** of the isolation structure **200** close to the other one of the two light-emitting units **310**. For example, the light-emitting units **310** include a blue light-emitting unit, a green light-emitting unit, and a red light-emitting unit, the fifth distance D**5** of the isolation structure **200** close to the blue light-emitting unit is greater than the fifth distance D**5** of the isolation structure **200** close to the green light-emitting unit, and the fifth distance D**5** of the isolation structure **200** close to the green light-emitting unit is greater than the fifth distance D**5** of the isolation structure **200** close to the red light-emitting unit, and thus it may be determined that the red light-emitting unit is prepared first, followed by the green light-emitting unit, and finally the blue light-emitting unit.
[0083] In some optional embodiments, the fifth distances D**5** close to the light-emitting units **310** of a same color are the same.
[0084] In these optional embodiments, the fifth distances D**5** of the isolation structures **200** located at the periphery of the light-emitting units **310** of a same color are the same, it may be determined that the light-emitting units **310** of a same color are prepared in a same step. Moreover, the fifth distances D**5** corresponding to the light-emitting units **310** of a same color are the same, and if the fifth distances D**5** corresponding to at least two light-emitting units **310** are different, the colors of lights emitted by the two light-emitting units **310** are different, and the two light-emitting units **310** are vapor deposited in different steps. Therefore, the fifth distances D**5** corresponding to the light-emitting units **310** of a same color being controlled to be the same facilitates determining the vapor deposition order for the light-emitting units **310** according to the fifth distances D**5** corresponding to the light-emitting units **310**.
[0085] Optionally, the light-emitting layer **300** includes an electron injection layer (EIL), an electron transport layer (ETL), a light-emitting material layer, a hole injection layer (HIL), and a hole transport layer (HTL).
[0086] Reference is made to FIG. **1**, the embodiments of the second aspect of the present application provide a display panel **10**, including: a base plate **100**, a plurality of isolation structures **200**, and a light-emitting layer **300**; the isolation structures **200** being located on the base plate **10**, the isolation structures **200** encircling a plurality of isolation openings **240**, the isolation structure **200** including a first layer **210** and a second layer **220** located at a side of the first layer **210** away from the base plate **100**, an orthographic projection of the first layer **210** on the base plate **100** being located within an orthographic projection of the second layer **220** on the base plate **100**; and the light-emitting layer **300** being located on the base plate **100** and including light-emitting units **310** located in the isolation openings **240**; in which the first layer **210** includes a top surface **211** at a side away from the base plate **100**, the second layer **220** includes a bottom surface

221 at a side close to the base plate **100**, the top surface **211** and the bottom surface **221** are in contact, a distance between an edge of an orthographic projection of the top surface **211** on the base plate **100** close to the light-emitting unit **310** and an edge of an orthographic projection of the bottom surface **221** on the base plate **100** close to the light-emitting unit **310** is an extension distance DO, and in at least two of the light-emitting units **310**, the extension distance D**0** of the isolation structure **200** close to one of the two light-emitting units **310** is greater than the extension distance D**0** of the isolation structure **200** close to the other one of the two light-emitting units **310**.
[0087] The display panel **10** according to the embodiments of the present application includes the base plate **100**, the isolation structures **200**, and the light-emitting layer **300**. The first layer **210** and the second layer **220** are arranged to form the isolation structure **200**, the orthographic projection of the first layer **210** arranged close to the base plate **100** on the base plate **100** is located within the orthographic projection of the second layer **220** on the base plate **100**, the area of the second layer **220** is greater than the area of the first layer **210**, the second layer **200** covers a surface of the first layer **210** close to the second layer **220**, and in such cases, the first layer **210** is recessed with respect to the second layer **220** in a direction away from the isolation opening **240**. When the light-emitting layer **300** is prepared, the light-emitting layer **300** has a large drop at the edge of the isolation structure **200**, and the first layer **210** is concave with respect to the second layer **220**, it's difficult for the light-emitting layer **300** to be connected at the edge of the isolation structure **200**, and thus fracture occurs and the light-emitting layer **300** is fractured, resulting in the light-emitting units **310** that are spaced apart from each other. In this way, the crosstalk of carriers within the light-emitting layer **300** is reduced, the display effect of the display panel **10** is improved, and the light-emitting units **310** can be prepared without a precision mask plate, so as to reduce the development and use of the precision mask plate and reduce the preparation cost. In at least two of the light-emitting units **310**, the extension distances D**0** of the isolation structure **200** close to the two light-emitting units **310** are different, and the vapor deposition order for the two light-emitting units **310** may be determined according to the different extension distances D**0** corresponding to the two light-emitting units **310**, i.e., according to the individual design of the isolation structure **200**, so as to obtain a display panel **10** in which the vapor deposition order for the light-emitting units **310** can be determined. In the preparation process, the isolation structures **200** may be wet etched in a wet etching process when the light-emitting units **310** of different colors are prepared, i.e., the first layers **210** of the isolation structures **200** may be etched. Therefore, in order to reflect the difference in the etching of the first layers **210**, the distance between the edge of the orthographic projection of the top surface **211** on the base plate **100** close to the light-emitting unit **310** and the edge of the orthographic projection of the bottom surface **221** on the base plate **100** close to the light-emitting unit **310** is used as the extension distance DO, so as to determine the vapor deposition order for the light-emitting units **310** according to the extension distances DO corresponding to the light-emitting units **310**.
[0088] Reference is made to FIG. **8**, which shows a partial sectional view of a display panel in yet another embodiment.
[0089] As shown in FIG. **8**, the embodiments of the third aspect of the present application provide a display panel **10**, including: a base plate **100**; a plurality of isolation structures **200** located on the base plate **100**, the isolation structures **200** encircling a plurality of isolation openings **240**, the isolation structure **200** including a first layer **210** and a second layer **220** located at a side of the first layer **210** away from the base plate **100**, an orthographic projection of the first layer **210** on the base plate **100** being located within an orthographic projection of the second layer **220** on the base plate **100**; and a light-emitting layer **300** located on the base plate **100** and including light-emitting units **310** located in the isolation openings **240**; in which a distance between orthographic projections of the first layers **210** located at two sides of a same isolation opening **240** on the base plate **100** is a sixth distance D**6**, and in at least two of the light-emitting units **310**, the sixth distance D**6** of the isolation structure **200** close to one of the two light-emitting units **310** is greater

than the sixth distance D**6** of the isolation structure **200** close to the other one of the two light-emitting units **310**.

[0090] The display panel **10** according to the embodiments of the present application includes the base plate **100**, the isolation structures **200**, and the light-emitting layer **300**. The first layer **210** and the second layer **220** are arranged to form the isolation structure **200**, the orthographic projection of the first layer **210** arranged close to the base plate **100** on the base plate **100** is located within the orthographic projection of the second layer **220** on the base plate **100**, the area of the second layer **220** is greater than the area of the first layer **210**, the second layer **200** covers a surface of the first layer **210** close to the second layer **220**, and in such cases, the first layer **210** is recessed with respect to the second layer **220** in a direction away from the isolation opening **240**. When the light-emitting layer **300** is prepared, the light-emitting layer **300** has a large drop at the edge of the isolation structure **200**, and the first layer **210** is concave with respect to the second layer **220**, it's difficult for the light-emitting layer **300** to be connected at the edge of the isolation structure **200**, and thus fracture occurs and the light-emitting layer **300** is fractured, resulting in the light-emitting units **310** that are spaced apart from each other. In this way, the crosstalk of carriers within the light-emitting layer **300** is reduced, the display effect of the display panel **10** is improved, and the light-emitting units **310** can be prepared without a precision mask plate, so as to reduce the development and use of the precision mask plate and reduce the preparation cost.

[0091] For example, the light-emitting units **310** include a blue light-emitting unit, a green light-emitting unit, and a red light-emitting unit, the sixth distance D**6** of the isolation structure **200** close to the blue light-emitting unit is greater than the sixth distance D**6** of the isolation structure **200** close to the green light-emitting unit, and the sixth distance D**6** of the isolation structure **200** close to the green light-emitting unit is greater than the sixth distance D**6** of the isolation structure **200** close to the red light-emitting unit, and thus it may be determined that the red light-emitting unit is prepared first, followed by the green light-emitting unit, and finally the blue light-emitting unit.

[0092] In some optional embodiments, the sixth distances D**6** close to the light-emitting units **310** of a same color are the same.

[0093] In these optional embodiments, the sixth distances D**6** of the isolation structures **200** located at the periphery of the light-emitting units **310** of a same color are the same, it may be determined that the light-emitting units **310** of a same color are prepared in a same step. Moreover, the sixth distances D**6** corresponding to the light-emitting units **310** of a same color are the same, and if the sixth distances D**6** corresponding to at least two light-emitting units **310** are different, the colors of lights emitted by the two light-emitting units **310** are different, and the two light-emitting units **310** are vapor deposited in different steps. Therefore, the sixth distances D**6** corresponding to the light-emitting units **310** of a same color being controlled to be the same facilitates determining the vapor deposition order for the light-emitting units **310** according to the sixth distances D**6** corresponding to the light-emitting units **310**.

[0094] In some optional embodiments, the sixth distances D**6** close to the light-emitting units **310** of different colors are different.

[0095] In these optional embodiments, the sixth distances D**6** of the isolation structures **200** located at the periphery of the light-emitting units **310** of different colors are different, it may be determined that the light-emitting units **310** of different colors are prepared in different steps. The sixth distances D**6** corresponding to the light-emitting units **310** of a same color are the same, and the sixth distances D**6** corresponding to the light-emitting units **310** of different colors are different, therefore the vapor deposition order for the light-emitting units **310** can be determined according to the sixth distances D**6** corresponding to the light-emitting units **310**.

[0096] Reference is made to FIGS. **9** to **11**, in which FIG. **9** shows a partial sectional view of a display panel in yet another embodiment, FIG. **10** shows a partial sectional view of a display panel in yet another embodiment, and FIG. **11** shows a partial top view of a display panel in yet another embodiment.

[0097] As shown in FIGS. **9** to **11**, according to any of the above implementation of the first aspect of the present application, the light-emitting units **310** include a first light-emitting unit **311**, a second light-emitting unit **312**, and a third light-emitting unit **313** of different colors, the sixth distance D**6** of the isolation structure **200** close to the first light-emitting unit **311** is a seventh distance D**7**, the sixth distance D**6** of the isolation structure **200** close to the second light-emitting unit **312** is an eighth distance D**8**, the sixth distance D**6** of the isolation structure **200** close to the third light-emitting unit **313** is a ninth distance D**9**, and at least two of the seventh distance D**7**, the eighth distance D**8**, and the ninth distance D**9** are different.
[0098] In these optional embodiments, at least two of the seventh distance D**7**, the eighth distance D**8**, and the ninth distance D**9** are different, i.e., the sixth distances D**6** corresponding to at least two of the first light-emitting unit **311**, the second light-emitting unit **312**, and the third light-emitting unit **313** are different, and the vapor deposition order for the light-emitting units may be determined according to the different sixth distances D**6**. For example, the ninth distance D**9** is greater than the eighth distance D**8**, i.e., the sixth distance D**6** of the isolation structure **200** close to the third light-emitting unit **313** is greater than the sixth distance D**6** of the isolation structure **200** close to the second light-emitting unit **312**, it may be determined that the second light-emitting unit **312** is prepared first, followed by the third light-emitting unit **313**; the eighth distance D**8** is greater than the seventh distance D**7**, i.e., the sixth distance D**6** of the isolation structure **200** close to the second light-emitting unit **312** is greater than the sixth distance D**6** of the isolation structure **200** close to the first light-emitting unit **311**, it may be determined that the first light-emitting unit **311** is prepared first, followed by the second light-emitting unit **312**. The structural designs in the embodiment may be applied to other display panels **10** and selected according to actual situation, which is not limited herein.
[0099] The embodiments of the fourth aspect of the present application further provide a display apparatus including the display panel **10** of any of the above embodiments. Since the display apparatus according to the embodiments of the fourth aspect of the present application includes the display panel **10** of any of the above embodiments, the display apparatus has the same beneficial effects as those of the display panel **10**, which will not be repeated herein.
[0100] The display apparatus in the embodiments of the present application includes, but is not limited to, a cellular phone, a Personal Digital Assistant (PDA), a tablet computer, an e-book, a television, an entrance guard, a smart fixed-line telephone, a console, and other apparatus with display function.
[0101] The embodiments of the fifth aspect of the present application further provide a method for manufacturing a display panel **10**, which may be the display panel **10** according to any of the above embodiments of the first aspect. Reference is made to FIGS. **12** to **16**, as well as FIGS. **1** to **11**, in which FIG. **12** shows a schematic flow chart of a method for manufacturing a display panel according to an embodiment of the present application, and FIGS. **13** to **16** show diagrams of a process for manufacturing a display panel according to an embodiment of the present application. The method includes the following steps: [0102] step S**01**: preparing a plurality of isolation structures on a base plate, a first isolation opening and a second isolation opening encircled by the isolation structures, the isolation structure including a first layer and a second layer located at a side of the first layer away from the base plate, an orthographic projection of the first layer on the base plate being located within an orthographic projection of the second layer on the base plate, a distance between an edge of the orthographic projection of the first layer on the base plate close to a light-emitting unit and an edge of the orthographic projection of the second layer on the base plate close to the light-emitting unit being an extension distance; [0103] step S**02**: preparing a first light-emitting material layer and a first electrode material layer on the base plate and patterning the first light-emitting material layer and the first electrode material layer to form a first light-emitting unit and a first sub-electrode located in the first isolation opening, the extension distance including a first distance close to the first light-emitting unit; [0104] step S**03**: preparing a second light-

emitting material layer and a second electrode material layer on the base plate and patterning the second light-emitting material layer and the second electrode material layer to form a second light-emitting unit and a second sub-electrode located in the second isolation opening, the extension distance including a second distance close to the second light-emitting unit, and the second distance being greater than the first distance.

[0105] In the method according to the embodiments of the fifth aspect of the present application, as shown in FIG. **7**, the isolation structure **200** is prepared in step S**01**, the first layer **210** and the second layer **220** are arranged to form the isolation structure **200**, and the distance between the edge of the orthographic projection of the first layer **210** on the base plate **100** close to a light-emitting unit **310** and the edge of the orthographic projection of the second layer **220** on the base plate **100** close to the light-emitting unit **310** is used as the extension distance D**0** of the isolation structure **200**. As shown in FIG. **8**, the first light-emitting material layer and the first electrode **410** material layer are prepared in step S**02**, and portions of the first light-emitting material layer and the first electrode **410** material layer located outside the first isolation opening **241** are etched, so as to obtain the first light-emitting unit **311** and the first sub-electrode **411** located in the first isolation opening **241**. As shown in FIG. **9**, the second light-emitting material layer and the second electrode material layer are prepared in step S**03**, and portions of the second light-emitting material layer and the second electrode material layer located outside the second isolation opening **242** are etched, so as to obtain the second light-emitting unit **312** and the second sub-electrode **412** located in the second isolation opening **242**. When the first electrode **410** material layer is etched, the first layer **210** of the isolation structure **200** at the periphery of the second isolation opening **242** is also etched, so that the second distance D**2** is greater than the first distance D**1**, i.e., the extension distance D**0** of the isolation structure **200** close to the second light-emitting unit **312** is greater than the extension distance DO of the isolation structure **200** close to the first light-emitting unit **311**. According to the first distance D**1** and the second distance D**2** which are different, the vapor deposition order for the first light-emitting unit **311** and the second light-emitting unit **312** may be determined.

[0106] As shown in FIG. **10**, in some optional embodiments, the isolation structures **200** further encircle a third isolation opening **243**, the method further includes, after step S**03**: preparing a third light-emitting material layer and a third electrode material layer on the base plate **100** and patterning the third light-emitting material layer and the third electrode material layer to form a third light-emitting unit **313** and a third sub-electrode **413** located in the third isolation opening **243**, the extension distance D**0** including a third distance D**3** close to the third light-emitting unit **313**, and the third distance D**3** being greater than the second distance D**2**.

[0107] In these optional embodiments, the third light-emitting material layer and the third electrode material layer are prepared, and portions of the third light-emitting material layer and the third electrode material layer located outside the third isolation opening **243** are etched, so as to obtain the third light-emitting unit **313** and the third sub-electrode **413** located in the third isolation opening **243**. When the second electrode material layer is etched, the first layer **210** of the isolation structure **200** at the periphery of the third isolation opening **243** is also etched, so that the third distance D**3** is greater than the second distance D**2**, i.e., the extension distance DO of the isolation structure **200** close to the third light-emitting unit **313** is greater than the extension distance D**0** of the isolation structure **200** close to the second light-emitting unit **312**. According to the second distance D**2** and the third distance D**3** which are different, the vapor deposition order for the second light-emitting unit **312** and the third light-emitting unit **313** may be determined.

[0108] Optionally, in the step of preparing the isolation structures **200** on the base plate **100**, the extension distances D**0** towards the first isolation opening **241**, the second isolation opening **242**, and the third isolation opening **243** are all the same, and before preparing the light-emitting units **310**, it is ensured that the extension distances D**0** corresponding to the isolation openings **240** are the same, so that when the extension distances D**0** corresponding to the isolation openings **240** differ from each other in a subsequent process, the vapor deposition order for the light-emitting units **310** may be determined according to the difference of the extension distances DO.

[0109] The above embodiments of the present application do not exhaustively describe all the details, nor do they limit the present application to the specific embodiments as described. Obviously, according to the above description, many modifications and changes can be made. These embodiments are selected and particularly described in the specification to better explain the principles and practical applications of the present application, so that a person skilled in the art is able to utilize the present application and make modifications based on the present application. The present application is limited only by the claims and the full scope and equivalents of the claims.

Claims

1. A display panel, comprising: a base plate; a plurality of isolation structures located on the base plate, the isolation structure comprising a first layer and a second layer located at a side of the first layer away from the base plate, an orthographic projection of the first layer on the base plate being located within an orthographic projection of the second layer on the base plate; a plurality of isolation openings encircled by the isolation structures; and a light-emitting layer located on the base plate and comprising light-emitting units located in the isolation openings; wherein a distance between an edge of the orthographic projection of the first layer on the base plate close to the light-emitting unit and an edge of the orthographic projection of the second layer on the base plate close to the light-emitting unit is an extension distance, and in at least two of the light-emitting units, the extension distance of the isolation structure close to one of the two light-emitting units is greater than the extension distance of the isolation structure close to the other one of the two light-emitting units.

2. The display panel according to claim 1, wherein the extension distances close to the light-emitting units of a same color are same; the extension distances close to the light-emitting units of different colors are different.

3. The display panel according to claim 1, wherein the first layer comprises a top surface at a side away from the base plate, the second layer comprises a bottom surface at a side close to the base plate, the top surface and the bottom surface are in contact, a distance between an edge of an orthographic projection of the top surface on the base plate close to the light-emitting unit and an edge of an orthographic projection of the bottom surface on the base plate close to the light-emitting unit is the extension distance.

4. The display panel according to claim 1, wherein the light-emitting units comprise a first light-emitting unit, a second light-emitting unit, and a third light-emitting unit of different colors, the extension distance of the isolation structure close to the first light-emitting unit is a first distance, the extension distance of the isolation structure close to the second light-emitting unit is a second distance, the extension distance of the isolation structure close to the third light-emitting unit is a third distance, and at least two of the first distance, the second distance, and the third distance are different.

5. The display panel according to claim 1, wherein the third distance is greater than the second distance; the second distance is greater than the first distance; and the first light-emitting unit is a light-emitting unit emitting red light, the second light-emitting unit is a light-emitting unit emitting green light, and the third light-emitting unit is a light-emitting unit emitting blue light.

6. The display panel according to claim 1, wherein the extension distance is in a range of 0.1 μm to 1.5 μm.

7. The display panel according to claim 1, wherein the display panel further comprises a pixel defining layer located on the base plate, the pixel defining layer comprises a plurality of pixel defining portions and a plurality of pixel openings encircled by the pixel defining portions, the pixel opening is communicated with the isolation opening; the isolation structure is located at a

side of the pixel defining portion away from the base plate.
8. The display panel according to claim 7, wherein the display panel further comprises a plurality of pixel electrodes located at a side of the pixel defining portions close to the base plate, an orthographic projection of the pixel electrode on the base plate at least partially overlaps an orthographic projection of the pixel opening on the base plate.
9. The display panel according to claim 7, wherein a distance between the edge of the orthographic projection of the first layer on the base plate close to the light-emitting unit and an edge of the orthographic projection of the pixel opening on the base plate is a fourth distance, and in at least two of the light-emitting units, the fourth distance of the isolation structure close to one of the two light-emitting units is greater than the fourth distance of the isolation structure close to the other one of the two light-emitting units; and the fourth distances close to the light-emitting units of a same color are the same, and the fourth distances close to the light-emitting units of different colors are different.
10. The display panel according to claim 1, wherein the display panel further comprises a first electrode layer located a side of the light-emitting layer away from the base plate; the first electrode layer comprises a plurality of first electrodes arranged at intervals, the first electrode is electrically connected with the isolation structure; an orthographic projection of the light-emitting unit on the base plate is located within an orthographic projection of the first electrode on the base plate.
11. The display panel according to claim 1, wherein the light-emitting units and the isolation structures are arranged alternately; the second layer comprises an electrically conductive material or an electrically insulating material.
12. The display panel according to claim 1, wherein the second layer comprises a metal material, the first layer and the second layer are of different materials.
13. The display panel according to claim 1, wherein the isolation structure further comprises a third layer located at a side of the first layer towards the base plate, the orthographic projection of the first layer on the base plate is located within an orthographic projection of the third layer on the base plate; a distance between an edge of the orthographic projection of the third layer on the base plate close to the light-emitting unit and the edge of the orthographic projection of the second layer on the base plate close to the light-emitting unit is a fifth distance, and in at least two of the light-emitting units, the fifth distance of the isolation structure close to one of the two light-emitting units is greater than the fifth distance of the isolation structure close to the other one of the two light-emitting units; and the fifth distances close to the light-emitting units of a same color are same.
14. A display panel, comprising: a base plate; a plurality of isolation structures located on the base plate the isolation structure comprising a first layer and a second layer located at a side of the first layer away from the base plate, an orthographic projection of the first layer on the base plate being located within an orthographic projection of the second layer on the base plate; a plurality of isolation openings encircled by the isolation structures; and a light-emitting layer located on the base plate and comprising light-emitting units located in the isolation openings; wherein the first layer comprises a top surface at a side away from the base plate, the second layer comprises a bottom surface at a side close to the base plate, the top surface and the bottom surface are in contact, a distance between an edge of an orthographic projection of the top surface on the base plate close to the light-emitting unit and an edge of an orthographic projection of the bottom surface on the base plate close to the light-emitting unit is an extension distance, and in at least two of the light-emitting units, the extension distance of the isolation structure close to one of the two light-emitting units is greater than the extension distance of the isolation structure close to the other one of the two light-emitting units, wherein the extension distances close to the light-emitting units of a same color are same; and the extension distances close to the light-emitting units of different colors are different.
15. The display panel according to claim 14, wherein the light-emitting units comprise a first light-

emitting unit, a second light-emitting unit, and a third light-emitting unit of different colors, the extension distance of the isolation structure close to the first light-emitting unit is a first distance, the extension distance of the isolation structure close to the second light-emitting unit is a second distance, the extension distance of the isolation structure close to the third light-emitting unit is a third distance, and at least two of the first distance, the second distance, and the third distance are different; the third distance is greater than the second distance; the second distance is greater than the first distance; and the first light-emitting unit is a light-emitting unit emitting red light, the second light-emitting unit is a light-emitting unit emitting green light, and the third light-emitting unit is a light-emitting unit emitting blue light.
16. A display panel, comprising: a base plate; a plurality of isolation structures located on the base plate, the isolation structure comprising a first layer and a second layer located at a side of the first layer away from the base plate, an orthographic projection of the first layer on the base plate being located within an orthographic projection of the second layer on the base plate; a plurality of isolation openings encircled by the isolation structures; and a light-emitting layer located on the base plate and comprising light-emitting units located in the isolation openings; wherein a distance between orthographic projections of the first layers located at two sides of a same isolation opening on the base plate is a sixth distance, and in at least two of the light-emitting units, the sixth distance of the isolation structure close to one of the two light-emitting units is greater than the sixth distance of the isolation structure close to the other one of the two light-emitting units.
17. The display panel according to claim 16, wherein the sixth distances close to the light-emitting units of a same color are same; and the sixth distances close to the light-emitting units of different colors are different.
18. The display panel according to claim 16, wherein the light-emitting units comprise a first light-emitting unit, a second light-emitting unit, and a third light-emitting unit of different colors, the sixth distance of the isolation structure close to the first light-emitting unit is a seventh distance, the sixth distance of the isolation structure close to the second light-emitting unit is an eighth distance, the sixth distance of the isolation structure close to the third light-emitting unit is a ninth distance, and at least two of the seventh distance, the eighth distance, and the ninth distance are different; the ninth distance is greater than the eighth distance; the eighth distance is greater than the seventh distance; and the first light-emitting unit is a light-emitting unit emitting red light, the second light-emitting unit is a light-emitting unit emitting green light, and the third light-emitting unit is a light-emitting unit emitting blue light.
19. A method for manufacturing a display panel, comprising: preparing a plurality of isolation structures on a base plate, a first isolation opening and a second isolation opening encircled by the isolation structures, the isolation structure comprising a first layer and a second layer located at a side of the first layer away from the base plate, an orthographic projection of the first layer on the base plate being located within an orthographic projection of the second layer on the base plate, a distance between an edge of the orthographic projection of the first layer on the base plate close to a light-emitting unit and an edge of the orthographic projection of the second layer on the base plate close to the light-emitting unit being an extension distance; preparing a first light-emitting material layer and a first electrode material layer on the base plate and patterning the first light-emitting material layer and the first electrode material layer to form a first light-emitting unit and a first sub-electrode located in the first isolation opening, the extension distance comprising a first distance close to the first light-emitting unit; and preparing a second light-emitting material layer and a second electrode material layer on the base plate and patterning the second light-emitting material layer and the second electrode material layer to form a second light-emitting unit and a second sub-electrode located in the second isolation opening, the extension distance comprising a second distance close to the second light-emitting unit, and the second distance being greater than the first distance.
20. The according to claim 19, wherein the isolation structures further encircle a third isolation

opening, the method further comprises, after forming the second light-emitting unit and the second sub-electrode located in the second isolation opening: preparing a third light-emitting material layer and a third electrode material layer on the base plate and patterning the third light-emitting material layer and the third electrode material layer to form a third light-emitting unit and a third sub-electrode located in the third isolation opening, the extension distance comprising a third distance close to the third light-emitting unit, and the third distance being greater than the second distance.
21. The according to claim 19, wherein in preparing the isolation structures on the base plate, the extension distances towards the first isolation opening, the second isolation opening, and the third isolation opening are all the same.
22. The according to claim 19, wherein the first light-emitting unit is a light-emitting unit emitting red light, the second light-emitting unit is a light-emitting unit emitting green light, and the third light-emitting unit is a light-emitting unit emitting blue light.